

Introduction to Removable Partial Prosthodontics

Learning objectives:

- Understand the fundamentals of removable partial dentures (RPDs)
- Learn the indications for RPDs
- Differentiate between RPDs and fixed partial dentures (FPDs)
- Classify RPDs based on support and major connector material
- Master the Kennedy Classification System

Removable partial denture is a prosthesis that replaces some teeth in a partially dentate arch and can be removed from the mouth and replaced at will.

Removable partial denture objectives:

1. Restore appearance, phonetics, and function.
2. Restore health, comfort, and quality of life.
3. Distribute occlusal forces favourably.
4. prevent the drifting of remaining teeth or over-erupting into the edentulous space.
5. Stimulation for the underlying ridge and mucosa (prevent disuse atrophy).

Indications of removable partial dentures

1. Distal extension situations (free end situation).
2. Long-span tooth-bounded edentulous area.
3. Excessive loss of the residual ridge.
4. If the prognosis of the remaining teeth is questionable.
5. After a recent extraction.
6. Economic consideration.



Fixed partial prosthesis (fixed bridge): A partial denture that is luted or otherwise securely retained to natural teeth, tooth roots, and /or dental implants that furnish the primary support and retention for the prosthesis.

Indications of a fixed partial denture

1. Unilateral bounded edentulous span.
2. Class IV Kennedy with normal bone loss.

Anterior Bridge

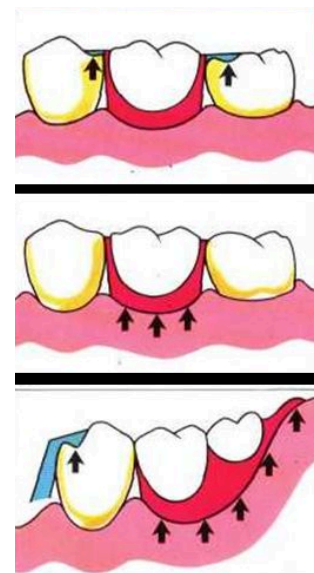


Classification of removable partial denture:

A. According to the support received

Support is the resistance to the movement of the denture toward the edentulous ridge.

1. Tooth supported
2. Tissue supported
3. Tooth and tissue-supported



B. According to the major connector material:

The major connector is that part of an RPD which connects components on one side of the arch to those on the other side. If the major connector is constructed of a cast metal alloy, the RPD is termed **definitive**. If the major connector is made of acrylic, the RPD is termed an **interim**.

1. **The definitive RPD:** is constructed after extensive diagnosis, treatment planning, and thorough preparation of the teeth and tissues for the prosthesis. The length of service of definitive RPD is intended to be many years.
2. **The interim RPD:** is usually constructed as part of the preparation of the mouth for definitive RPD, FPD or implant treatment. The length of service of the interim RPD is generally planned to be a year or less. They are frequently referred to as temporary RPD.



C. According to the arch configuration

The most widely accepted system of classification of RPD and partially edentulous arches was proposed by Dr Edward Kennedy in 1923. It is based on the configuration of the remaining natural teeth and edentulous spaces. This system was further defined and expanded upon by Dr O.C. Applegate and Dr Jacques Fiset.

The value of the Kennedy Applegate-Fiset classification system is that it is relatively simple, easy to remember, extremely comprehensive and very practical. It permits visualisation of the partially edentulous arch or RPD designed for that arch. It indicates the type of support for the RPD, which suggests certain physiologic and mechanical principles of treatment and RPD design. This system is widely taught and generally accepted, and used. The system also simplifies communication.

1. Class I: Bilateral posterior edentulous areas.
2. Class II: unilateral posterior edentulous area
3. Class III unilateral tooth-bounded edentulous area.
4. Class IV: single anterior tooth bounded edentulous area which crosses the midline.

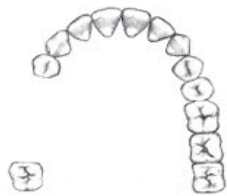
In Kennedy classification, a modification is any edentulous space (a gap where teeth are missing) other than the one that defines the main Kennedy class of the partial denture. Modifications are designated by their number (e.g., Modification 1, Modification 2), indicating how many additional edentulous areas exist, but the extent of the space does not matter.



Kennedy Class I



Kennedy Class II



Kennedy Class III

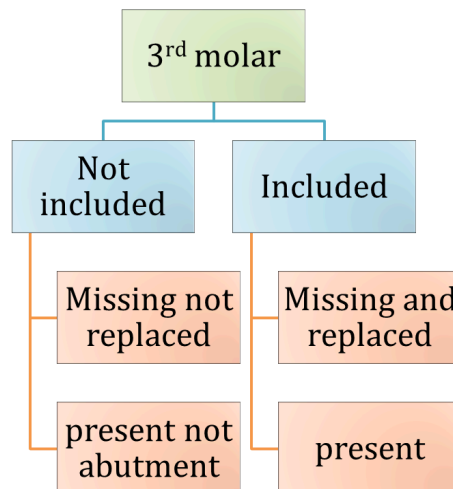


Kennedy Class IV

Applegate's Rules for Applying the Kennedy Classification

- Rule 1: The classification should follow, not precede extractions.
- Rule 2: If a 3rd molar is missing and not to be replaced, it is not considered in the classification.
- Rule 3: If a 3rd molar is present and not to be used as an abutment, it is not considered in the classification.
- Rule 4: If a 2nd molar is missing and not to be replaced, it is not considered in the classification.
- Rule 5: The most posterior area always determines the classification.

- Rule 6: Edentulous areas other than those determining the classification are referred to as modifications and designated by their No. e.g. class I, modification 1
- Rule 7: The extent of the modifications is not considered; only the number of additional edentulous areas.
- Rule 8: There are no modifications in Class IV.



Group		Number of missing teeth	Kennedy classification
Control		0	Not applicable
Case 1		2	Class III
Case 2		5	Class III Modification 1
Case 3		5	Class IV
Case 4		6	Class III Modification 1

*References

McCracken's Removable Partial Prosthodontics, 14th Edition.